Unit 6, Lesson 6

# Multiply Two-Digit Numbers and One-Digit Numbers

## Standards

Addressing **4.NBT.B.5**

## Instructional Routines

- Notice and Wonder

## Goals

- Comprehend (in spoken language) the meaning of the term "partial products."
- Interpret (orally and in writing) diagrams that represent partial products as a strategy for multiplying two-digit and one-digit whole numbers.
- Represent products of two-digit and one-digit numbers with a diagram and calculate the value.

## Student Facing Learning Goals

Let's multiply two-digit and one-digit numbers.

## Lesson Purpose

The purpose of this lesson is for students to multiply a two-digit number and a one-digit number using place value understanding.

## Narrative

In the previous lesson, students solved two-digit multiplication problems in a way that made sense to them. Students discussed decomposing factors and considered different representations of their strategy. In this lesson, students extend these ideas to find the value of products beyond 100, focusing on representations and strategies based on place value and the properties of operations, which are familiar from IM Grade 3.

Students analyze base-ten diagrams and diagrams that involve rectangles, some of which are partitioned by place value. Students explain how the diagrams represent multiplication and make connections between them, deepening their understanding of place value and properties of operations. At the end of the lesson, students consider a rectangular diagram that will be used through the rest of the section.

## Access for Students with Disabilities

- Engagement

## Access for English Learners

- MLR1

## Lesson Timeline

| | |
|---|---|
| Warm-up | 10 min |
| Activity 1 | 20 min |
| Activity 2 | 15 min |

## Teacher Reflection Questions

What strategy did most students use in their work today? What strategy did you anticipate today? Which did you not anticipate?

| | |
|---|---|
| Synthesis Estimate | 10 min |
| Cool-down | 5 min |

# Warm-up

10 min

Notice and Wonder: With and Without a Grid

## Standards

Addressing **4.NBT.B.5**

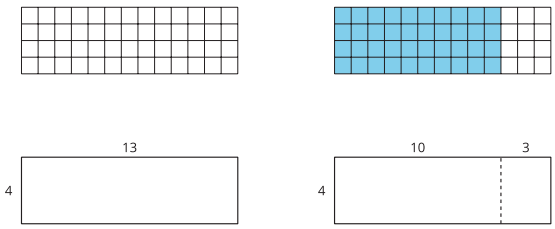
## Instructional Routines

- Notice and Wonder

The purpose of this *Warm-up* is to elicit students' prior knowledge of area and the idea that a rectangle can be decomposed into smaller rectangular regions. Students look at 4 different area diagrams they used in IM Grade 3. The reasoning will be useful when students use diagrams to multiply two- and one-digit numbers in a later activity. While students may notice and wonder many things about the number of units within the area of the gridded region, focus on the connections between the diagrams with a grid and those without.

## Student Task Statement

What do you notice? What do you wonder?




## Student Response

Students may notice:

- Two rectangles are gridded and have no numbers. The other two have no grid and have numbers.
- In the first two diagrams, there are 13 squares in each row and 4 squares in each column.
- A portion of the gridded rectangle is shaded.
- Some rectangles have the side lengths marked.
- The last rectangle is partitioned into two sections.

Students may wonder:

- Where are the numbers in the gridded rectangles?
- Why is the second rectangle shaded?
- Do the four diagrams represent the same rectangle?

## Launch

- Groups of 2
- Display the image.
- "What do you notice? What do you wonder?"
- 1 minute: quiet think time

## Activity

- "Discuss your thinking with your partner."
- 1 minute: partner discussion
- Share and record responses.

## Activity Synthesis

- "What do you know about the 13 and 4 in the first ungridded rectangle?" (They are side lengths. They correspond to the number of squares across and the number of squares down in the first rectangle.)
- "What about the 10, 3, and 4 in the second ungridded rectangle?" (The 10 and 3 are numbers that add up to 13. The 4 is the length of the shorter side.)
- "How are the four diagrams related? Name as many connections as you see." (They all represent the same rectangle because they have the same side lengths and the same area. The first two show the number of square units that fit in the rectangle. The last two don't show it but we can tell by multiplying the side lengths. The shaded portion in the second

gridded rectangles show the $4 \times 10$ portion in the fourth rectangle.)

- If no students mentioned the area of the rectangles in their analysis, ask: "How might we find the area of the rectangles?" (For the gridded rectangles, we can count the unit squares or multiply the number of units across and down. For the other two, we can multiply the side lengths.)

# Activity 1

20 min

Tyler's Diagrams

## Standards

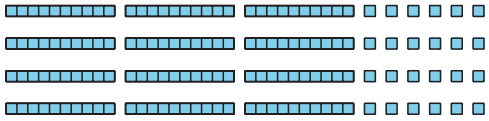
Addressing **4.NBT.B.5**

This activity prompts students to make sense of base-ten diagrams for representing multiplication. The representation supports students in grouping tens and ones and encourages them to use place value understanding and to apply the distributive property (MP7).

This activity is an opportunity for students to build conceptual understanding of partial products in a more concrete way. In the next activity, students will notice that working with these drawings can be cumbersome and transition to using rectangular diagrams, which are more abstract.

## Student Task Statement

1. Tyler uses this base-ten diagram to find the value of $4 \times 36$.

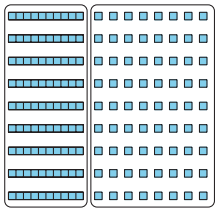


   a. Where is the 36 in Tyler's diagram?

   b. Where is the 4 in his diagram?

   c. What is the value of $4 \times 36$?

2. Tyler makes a diagram to find the value of $9 \times 18$.



   Explain or show how his diagram helps him find the value of $9 \times 18$.

## Launch

- Groups of 3–4

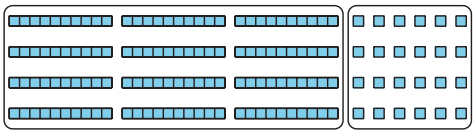
## Activity

- 5 minutes: independent work time on the first problem
- 2–3 minutes: partner discussion
- 3–4 minutes: independent work time on the second problem
- Monitor for students who use different methods to find the value of $4 \times 36$ in the first problem.

## Activity Synthesis

- Select 2 students who organized the diagram in the first problem in different ways and then compare their diagrams to Tyler's method.
- Invite selected students to share their method for finding the value of $4 \times 36$.

### Student Response

1. a. Each row has 3 tens and 6 ones.
   b. There are 4 rows of 3 tens and 6 ones.
   c. 144. Sample responses:
      - There are 12 tens and 24 ones, or 120 and 24, which make 144.
      - $(4 \times 30) + (4 \times 6) = 120 + 24$, which is 144.
      - 
2. Sample responses:
   - The 9 tens make 90, and the 9 groups of 8 ones make 72, so the product is $90 + 72$, which is 162.
   - $(9 \times 10) + (9 \times 8) = 90 + 72 = 162$

- "How are these methods the same? How are they different?"
- "How are these methods like Tyler's method to find the value of $9 \times 18$? How are they different?"

# Activity 2

15 min

Two Kinds of Diagrams

## Standards

Addressing **4.NBT.B.5**

This activity continues to encourage place value reasoning for finding the product of a two-digit factor and a one-digit factor.

Students make sense of two representations that show the two-digit factor decomposed by place value: a base-ten diagram and a rectangle. The latter looks like an area diagram that students have used in IM Grade 3, where the side lengths of a rectangle represent two factors. As the factors become larger, however, it becomes necessary to draw rectangles whose side lengths are not proportional. When rectangles no longer accurately represent area, the term "area diagrams" is not used. Instead, "rectangular diagrams" is used in teacher materials and "diagrams" in student materials.

Students then choose a representation to use to find products and write corresponding expressions. In the *Activity Synthesis*, they learn that the results of multiplying a part of one factor by the other factor can be called "partial products." In future lessons, students will use rectangular diagrams to represent multiplication of larger numbers.

## Access for English Language Learners

*MLR1 Stronger and Clearer Each Time.* Synthesis: Before the whole-class discussion, give students time to meet with 2–3 partners to share and get feedback on their response to which method they prefer when multiplying 6 by 53. Invite listeners to ask questions, to press for details, and to suggest mathematical language. Give students 2–3

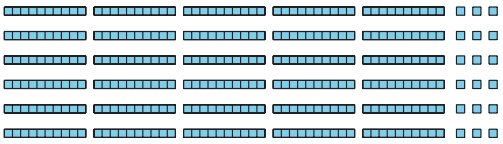
minutes to revise their written explanation based on the feedback they receive.
*Advances: Writing, Speaking, Listening*

## Access for Students with Disabilities

*Engagement: Provide Access by Recruiting Interest.* Optimize meaning and value. Invite students to share examples from their own lives in which they might need to multiply two-digit and one-digit numbers. Invite them to imagine and share why Han and Priya might be multiplying 6 times 53.
*Supports accessibility for: Attention, Social-Emotional Functioning*

## Student Task Statement

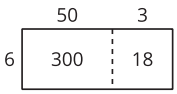
1. Priya draws a base-ten diagram to multiply $6 \times 53$. She says it shows that the product can be found by adding 300 and 18.



   a. Where do you see 6 and 53 in Priya's diagram?

   b. Where do you see 300 and 18 in her diagram? What do they represent?

2. Han draws this diagram to multiply $6 \times 53$:




   Where do you see 300 and 18 in his diagram? What do they represent?

3. Which diagram do you prefer for multiplying $6 \times 53$: Han's way or Priya's way? Explain your reasoning.

4. Find the value of $6 \times 53$.

5. Draw a diagram to represent each multiplication expression. Then find the value of each product.

   a. $6 \times 48$

   b. $9 \times 67$

## Student Response

1. Sample responses:
   - Each row has 5 tens and 3 ones, which represent 53. The six rows represent the 6.
   - The 300 is the 30 tens, which comes from 6

## Launch

- Groups of 2

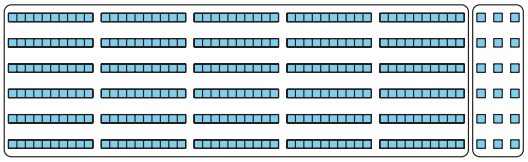
## Activity

- "Take a few quiet minutes to work on the activity. Afterward, share your thinking with your partner."
- 6–7 minutes: independent work time
- 3–4 minutes: partner discussion
- As students work on the last problem, monitor for those who:
  - Recognize that base-ten drawings are cumbersome and a less efficient way to find the product.
  - Draw diagrams that show the two-digit side length partitioned by tens and ones.
  - Write expressions that show place value reasoning.
  - Use the distributive property (find the partial products, then add to find the value of the product).

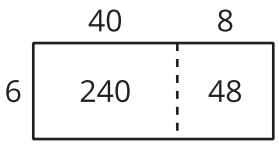
## Activity Synthesis

- Select 2–3 students to share their responses and reasoning.
- "How are Han's and Priya's diagrams similar?" (They both show the 53 as 50 and 3. They multiply the parts and add them to find the total value.)
- "How are they different?" (Han just drew a rectangle, and Priya drew all the tens and ones.)
- "What expressions can you write to show how to multiply $6 \times 53$? What is the product?" ($(6 \times 50) + (6 \times 3)$. The product is 318.)
- Explain that the 300 and 18 in both Han's and Priya's

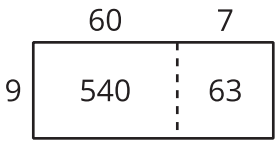
groups of 5 tens. The 18 is the 18 ones, which comes from 6 groups of 3.



2. Sample response: The 300 and 18 represent the areas of the rectangles if we had created a diagram like Priya's but is easier to find 300 is $6 \times 50$ and the 18 is $6 \times 3$.
3. Sample responses:
   - Han's way, because it is easier to see and multiply the numbers.
   - Priya's way, because it's easier to count the tens and ones.
4. $6 \times 53 = 318$
5. a. 288




   b. 603




- For the last problem, if students do not partition their rectangular diagrams by place value, display the sample diagrams as shown in the student responses without products. Ask:
  - "What expressions can we write to represent each part of the diagram?" ($6 \times 40$ and $6 \times 8$ or $9 \times 60$ and $9 \times 7$.)
  - "How does decomposing the two-digit number into tens and ones help to find the product?" (Finding multiples of ten and facts like $6 \times 8$ is easier than finding a product of say, 36 and 8. Then we can add the smaller products together to find the total product.)

diagrams are called partial products. Each is found by multiplying a part of one of the factors by the other factor.

## Lesson Synthesis

"Today we used different diagrams and expressions to represent multiplication."

Display $8 \times 79$ and a blank rectangle.



"How could we use this rectangle to represent $8 \times 79$?" (Place 8 on one of the sides of the rectangle and partition the rectangle to show 79 across the other side.)

Label the rectangle and partition the rectangles to show 79 decomposed into 70 and 9.

"How does this diagram help us find the value of $8 \times 79$?" ($8 \times 79$ is hard to do mentally, but I know that it is like finding the area of a rectangle with side lengths 8 and 79. I can decompose the rectangle into smaller rectangles and add the areas of the smaller rectangles to find the area of the large rectangle. Also, I can do $8 \times 70$ and $8 \times 9$ in my head, which makes the multiplication easier.)

"What is the value of the product?" (632, because $8 \times 70$ is 560 and $8 \times 9$ is 72, and $560 + 72 = 632$)

## Suggested Centers

- Can You Draw It? (1–5), Stage 4: Area and Perimeter (Supporting)
- Five in a Row: Multiplication (3–5), Stage 2: Factors 1–9 (Supporting)

# Cool-down

5 min

Represent the Product

## Standards

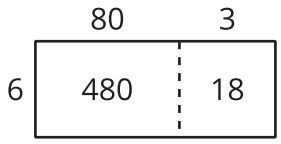
Addressing **4.NBT.B.5**

## Student Task Statement

Find the value of $6 \times 83$. Use a diagram if it is helpful.

## Student Response

Sample response:




$6 \times 80 = 480$
$6 \times 3 = 18$
$480 + 18 = 498$

## Responding to Student Thinking

Students use a diagram, but find a value other than 498 as the product.

Next Day Supports
Launch *Warm-up* or activities by highlighting important representations from previous lessons.